China AI Compute Capacity Study

Does China's Advanced Chip Production Actually Support Its AI Development Timeline?

China has made a series of high-profile announcements about the maturation of its domestic AI chip industry. Semiconductor Manufacturing International Corporation (SMIC) has confirmed volume production of 7nm-class chips without extreme ultraviolet (EUV) lithography. Huawei's Ascend accelerators have shipped in meaningful quantities to Chinese technology firms. Government statements speak of compute self-sufficiency as an achievable near-term objective. Beijing's investment posture — billions of dollars directed at fabs, memory, and chip design — has reinforced that narrative at every turn.

This memo asks a different question: do those programs actually produce usable chips at the scale a competitive AI development effort requires?

The distinction matters because announced capacity and realized capacity are not the same thing. A fab that can demonstrate a 7nm process node is not the same as a fab that can produce 7nm chips at the volumes, yields, and costs needed to supply a frontier AI research program. The gap between what Chinese industry can demonstrate and what it can deliver at scale is the operative variable for policymakers — and the public record, examined carefully, suggests that gap remains substantial. The ongoing black market for smuggled Nvidia chips is the market's own verdict on whether domestic production is adequate.

This memo draws on public data from industry analysts, export control proceedings, earnings disclosures, and semiconductor trade press to construct a realistic estimate of China's annual advanced AI chip output. That figure is then compared against the compute requirements of a frontier AI development program. The analysis closes with a set of leading indicators policymakers should monitor to detect when the gap is meaningfully closing.

What Frontier AI Development Actually Requires

Understanding what a competitive AI development program actually requires is the necessary starting point for evaluating what China can and cannot supply.

Training a frontier large language model comparable to the systems fielded by leading

U.S. laboratories requires, at current scales, somewhere between 20,000 and 100,000 high-performance AI accelerators running in a tightly coupled cluster for several months. Epoch AI analysis puts the compute for GPT-4 at approximately 2 x 10^{25} floating-point operations, at an estimated hardware cost of $80 to $100 million.[1] The frontier has moved considerably since then. By 2026, leading training runs sit between 10^{26} and 10^{27} FLOPs, with cost estimates for GPT-5-class systems in the range of $200 million to $500 million for compute alone.[2] xAI's Colossus cluster in Memphis grew from 100,000 to 200,000 H100-equivalent accelerators over the course of 2025. Meta's internal H100-class capacity exceeded 350,000 equivalents by the end of that year.[3]

Training and inference are not separable activities at this scale. The same compute that trains frontier models also serves them, meaning that a credible AI development program requires not a single large training cluster but a sustained, scalable supply of new chips to expand capacity and replace aging hardware.

The reference chip for this analysis is Nvidia's H100, which remains the working unit of frontier compute in 2026. Comparing Chinese output to U.S. capability requires accounting for the performance gap between domestically produced chips and the H100, not simply comparing unit counts.

SMIC's Process Technology: What the Node Announcements Mean

The headline finding from Chinese semiconductor development is genuine: SMIC has achieved production of chips using a 7nm-class process node without EUV lithography. TechInsights confirmed in December 2025 that the Huawei Kirin 9030 system-on-chip was manufactured on SMIC's N+3 node, a 5nm-class process, making it the most advanced chip produced in China without EUV tools.[4] The engineering involved is not trivial, and the achievement represents genuine progress against what many outside observers considered an insurmountable constraint.

What the announcement does not convey is the cost structure and yield performance that determine how many usable chips that process actually produces.

SMIC's advanced nodes rely on deep ultraviolet immersion (DUVi) lithography combined with multi-patterning techniques. A modern 7nm logic chip processed at TSMC requires approximately nine EUV lithography steps for its most critical layers. At SMIC, the absence of EUV forces the same features to be printed across 34 steps, using repeated exposures and etchings to achieve equivalent resolution.[5] The 5nm-class N+3 node requires four-exposure multi-patterning on its most critical metal layers. Every additional patterning step introduces opportunities for alignment error. Those errors compound, producing chips that fail to meet specification.

The result is a structurally lower yield rate than leading-edge fabs achieve with EUV. Yield is the share of chips on a wafer that meet performance specifications; it determines the effective cost per working chip. Multiple industry sources estimate SMIC's 7nm yield rate at below 50% as of mid-2024, with improvement to roughly 40% for the Ascend 910C by late 2024 and a target of 60% in 2025.[6] By contrast, TSMC's EUV-based processes at comparable nodes achieve yields that industry participants generally describe as industry-standard. The American Enterprise Institute estimated that SMIC's 5nm chip fabrication is 40 to 50 percent more expensive than TSMC's equivalent, with yields as low as 20 percent at earlier production stages.[7] The Financial Times quoted sources placing SMIC's 7nm yield at less than one-third of TSMC's comparable output.[8]

Lower yield has two practical consequences. First, each wafer produces fewer usable chips. If SMIC's 7nm process yields 40 usable dies per wafer where TSMC produces 90, the effective output of a given fab capacity is less than half. Second, cost per chip rises proportionally. Huawei and SMIC have reportedly priced advanced-node chips at a 40 to 50 percent premium over TSMC's equivalent,[9] a figure that reflects production economics rather than commercial positioning.

Breaking Down the Output Math

As of the third quarter of 2025, SMIC operated roughly 20,000 wafers per month of capacity at its most advanced nodes, with Huawei reportedly absorbing approximately 15,000 of those wafers monthly for Ascend production.[10] UBS analysis estimated China's combined advanced-node capacity at 30,000 to 50,000 wafer starts per month in 2025, projecting growth to 50,000 to 60,000 in 2026. The stated longer-term target, according to industry discussions cited in the same UBS note, is 150,000 to 160,000 wafers per month by end-2027 across multiple fabs.[11]

Even accepting the 2026 expansion figures, the usable chip output is considerably lower than raw wafer capacity suggests. A 12-inch wafer at the 7nm node produces roughly 50 to 100 die at full theoretical yield. At SMIC's demonstrated yield rates of 40 to 50 percent for its most advanced nodes, a working upper bound for usable chips is 25 to 50 per wafer. At 20,000 advanced-node wafer starts per month dedicated to Ascend, that ceiling is much lower than the raw wafer count implies — and it shrinks further once the full supply chain is examined.

The Memory Constraint: A Binding Bottleneck

High-bandwidth memory (HBM), stacked directly onto the compute die, determines an accelerator's effective throughput — and the HBM constraint on Chinese AI chip production is currently more binding than the logic constraint. Without adequate HBM supply, compute dies cannot reach their theoretical performance ceiling regardless of how many wafers SMIC processes.

Huawei built a substantial stockpile of HBM from Samsung before U.S. export restrictions on HBM sales to China took effect in late 2024. SemiAnalysis estimated this stockpile at approximately 11.7 million units, with 7 million shipped in a single month by Samsung before the restrictions were enforced. That stockpile supported Ascend production through 2025, but was expected to be substantially depleted by year-end.[12]

China's domestic answer is ChangXin Memory Technologies (CXMT), its leading DRAM producer. CXMT has achieved HBM3 sample production, drawing on government funding, foreign equipment procured before restrictions took effect, and reportedly engineers recruited from overseas firms, but its output remains well short of what Huawei's accelerator program requires. Industry projections estimate CXMT's output at approximately 2.2 million HBM stacks in 2026, with initial yields around 50 percent.[13] At that production rate, CXMT's output would support between 250,000 and 400,000 Ascend 910C packages — considerably below Huawei's stated production ambitions of 600,000 units of the 910C in 2026.[14] CXMT has not yet entered HBM3E mass production, which is required for the next generation of Huawei's accelerator roadmap. South Korean competitors are already moving to HBM4.[15]

The HBM constraint effectively caps China's realistic advanced AI chip output well below what its fab capacity could theoretically support. Even if SMIC doubles its advanced-node wafer output, the chips cannot be assembled into functional accelerators without matched HBM supply.

Comparing Output to Frontier Requirements

Working from the figures above, a realistic estimate of China's domestically sourced advanced AI chip output in 2025 sits at roughly 400,000 to 700,000 Ascend units across the 910B and 910C families, with a significant portion of that figure dependent on TSMC-fabricated die from a pre-sanctions stockpile that multiple analysts expected to run out by the end of 2025.[16]

Huawei's official 2024 figures confirm the challenge. The company shipped 200,000 Ascend 910B units that year, while Nvidia sold approximately one million H20 chips into China in the same period.[17] Huawei's chips are also performance-constrained relative to the H100: the Ascend 910C delivers roughly 60 to 80 percent of H100 inference performance and no better than that for training workloads, which are more sensitive to interconnect bandwidth and memory hierarchy than inference tasks.[18] In H100-equivalent terms, China's domestic supply in 2025 was plausibly in the range of 200,000 to 400,000 H100-equivalents per year.

A U.S. frontier AI program deploying 100,000 H100s in a single cluster — the baseline for competitive training in 2026 — consumes in one installation what China's entire domestic production could supply in six months to a year, and it does so at a higher performance level than the domestically produced equivalent. Maintaining, expanding, and refreshing that cluster requires a sustained pipeline that Chinese domestic production cannot currently match. The math does not support the claim that China's domestic chip industry can today supply a frontier AI development program at U.S. scale.

The Smuggling Market as Confirmation

The persistence and scale of the illegal market for Nvidia chips reflect a supply gap large enough to make significant legal and financial risk worthwhile for a broad range of actors.

Operation Gatekeeper, unsealed by the Department of Justice in December 2025, exposed a smuggling network that exported or attempted to export at least $160 million worth of Nvidia H100 and H200 GPUs to China between October 2024 and May 2025, with operatives using shell companies and falsified shipping paperwork to conceal the chips' ultimate destination.[19] A separate indictment unsealed in March 2026 charged that a scheme involving a former Supermicro co-founder had attempted to route AI servers worth $2.5 billion to Chinese customers through a front company in Southeast Asia, with shipments escalating to $150 million in a single two-month window in spring 2025.[20] A November 2025 case described a Florida-based front company that received $4 million in wire transfers from Chinese firms to purchase and export Nvidia chips.[21] These are not isolated incidents; they are a pattern, and enforcement officials have suggested the cases disclosed so far may represent the visible fraction of a larger flow.

SemiAnalysis analyst Ray Wang estimated that more than 60 percent of leading AI models developed in China currently run on Nvidia hardware.[22] That figure, if accurate, is the most direct statement available about the adequacy of domestic production. Chinese AI developers are not turning to smuggled foreign chips out of preference; they are doing so because those chips represent a generational performance advantage that domestic equivalents have not closed.

The software efficiency strategies that Chinese labs have developed under hardware scarcity — most notably the mixture-of-experts architecture and inference optimization techniques associated with DeepSeek's V3 and R1 models — are genuine innovations. They demonstrate that talented engineers can accomplish meaningful things under constrained compute budgets. They do not, however, eliminate the compute requirements of frontier training. A more efficient training recipe reduces the chips needed for a given model quality; it does not make a 100,000-chip training cluster possible with 10,000 chips.

The Optimistic Read and Its Limits

The most credible version of the optimistic case for China's AI chip self-sufficiency rests on three developments: the confirmed achievement of 5nm-class production at SMIC, the improvement in Ascend 910C yield rates from 20 to 40 percent over the course of 2024, and the rapid scaling of CXMT's HBM capacity. All three are real advances. None has yet translated into productive output at the scale a frontier AI program requires.

On the logic side, yield improvement from 20 to 40 percent is significant, but 40 percent remains well below the 60 percent threshold that Huawei and SMIC themselves identify as industry standard for profitable production.[23] At 40 percent yield, a wafer that could theoretically produce 80 functional dies produces 32. The cost and throughput implications of that gap compound across large production volumes. Reaching 60 percent yield at SMIC's advanced nodes — using multi-patterning techniques that EUV-equipped fabs do not require — involves a considerably steeper path than yield improvement at a mature EUV-based process, because the root causes of yield loss are more numerous and harder to isolate.

On the memory side, CXMT's HBM3 mass production schedule has slipped repeatedly. As of mid-2026, industry sources described its HBM3 supply as still in testing and sample-production phases, with mass production timelines uncertain.[24] The distance between CXMT's current output and what Huawei needs to scale Ascend production to a million units annually remains considerable. And beyond HBM3, the path to HBM3E and HBM4

— the memory generations used in leading accelerators today and tomorrow — faces the same EUV-access constraints that limit progress on the logic side.[25]

The AEI analysis of China's DUV machine stockpile offers perhaps the most informed view of the longer-term ceiling. Ninety DUV machines acquired in 2024 could theoretically support several million 7nm-class wafer starts. But that estimate assumes that multi-patterning yields improve substantially and that the software tools needed to configure DUV machines for cutting-edge multi-patterning remain accessible.[26] The December 2024 Bureau of Industry and Security rule specifically targeted the latter, restricting software that enables DUV machines to be configured for advanced multi-patterning. That restriction is a meaningful constraint on SMIC's ability to further improve its advanced-node process.

Leading Indicators for Policymakers

The gap between announced capability and realized output is substantial, but it is not static. Policymakers should monitor the following developments as signals that it is meaningfully closing.

Yield Rate Disclosures. SMIC does not publicly disclose advanced-node yield rates, but estimates surface through supply chain sources, earnings commentary, and trade press. A sustained improvement of SMIC's 5nm yield above 50 percent — particularly if accompanied by evidence that throughput is rising in proportion to stated capacity — would indicate a genuine step change in productive output rather than incremental improvement.

CXMT HBM Mass Production. The HBM bottleneck is currently the most binding constraint on Chinese AI chip supply. CXMT's transition from sample production to genuine mass production of HBM3 at scale, with yields sufficient to supply hundreds of thousands of Ascend packages, would represent a qualitative change in China's AI hardware position. Monitoring CXMT's customer deliveries, IPO disclosures, and production equipment purchases is a reasonable proxy.

Domestic EUV Development. Multiple Chinese institutions are pursuing domestic EUV lithography. A Chinese EUV prototype reportedly entered testing in late 2025.[27] Commercial-grade EUV capable of production use would substantially change the yield economics of advanced-node production and remove the constraint that is most fundamental to the current gap. The timeline for this development remains highly uncertain; the gap between a prototype and a production-capable tool is historically very large.

Smuggling Volume and Enforcement Patterns. If the black market for Nvidia chips contracts despite continued enforcement pressure, that could indicate that domestic alternatives are becoming adequate substitutes. If it expands or shifts to even more advanced hardware despite rising legal risk, that indicates the gap is widening rather than closing.

Software Stack Maturation. China's AI chips are competitive with Nvidia's H100 on raw compute figures much more than on the software ecosystem that makes those chips efficient to use. The CUDA software platform that Nvidia has built over two decades enables developers to extract near-theoretical throughput from H100 clusters. Huawei's CANN software stack and the broader Ascend ecosystem are improving but not yet comparable. Tracking adoption of Ascend hardware by leading Chinese AI laboratories — relative to continued reliance on smuggled or grey-market Nvidia chips — is an indicator of whether the software gap is closing.

Policy Implications

That export controls are working better than a pessimistic reading of Chinese announcements would suggest is not grounds for complacency; it is grounds for ensuring that the controls currently in place remain targeted at the constraints that actually matter.

The controls that matter most are those targeting the chokepoints this memo has identified: EUV lithography access, HBM memory supply, and the software tools that enable multi-patterning on DUV machines. The December 2024 BIS rule addressed the last of these. Sustained attention to HBM supply chains — including the intermediaries through which restricted memory reaches Chinese firms — addresses the second. The first remains contingent on maintaining allied consensus behind existing restrictions on ASML's most advanced equipment.

Enforcement of existing controls is not a substitute for updating them as technology evolves. As CXMT's HBM production scales, as SMIC pushes toward 3nm-class multi-patterning, and as Huawei's software ecosystem matures, the specific technologies that define the chokepoints will shift. Intelligence collection and analysis focused on the actual output of China's advanced chip programs — not just their announced capabilities — should inform decisions about where controls need to be tightened, maintained, or reconsidered.

The productive gap documented here is real and, under current conditions, consequential. It is also narrowing, and the pace of that change depends in large part on decisions made now about which technologies to restrict, which supply chains to monitor, and which leading indicators to treat as thresholds for policy action rather than data points to observe passively.

Notes

[1] Stanford University Human-Centered AI Institute, *AI Index Report 2025* (2025); see also Epoch AI estimates cited in Galileo AI, "How Much Does LLM Training Cost?" (2026).

[2] Deluair Consultancy, "Frontier AI Training Cost Trajectory 2026: The Run Rate, the Deal Stack, and the Power-Bound Horizon" (2026).

[3] Deluair Consultancy (2026); Meta capital expenditure disclosures, 2025.

[4] TechPowerUp, "Chinese SMIC Achieves 5nm Production on N+3 Node Without EUV Tools" (December 11, 2025); TechInsights analysis of Huawei Kirin 9030.

[5] Andy Lin, "How is SMIC After US Embargo?" *Granite Firm* (March 2025) (citing industry sources on the 9-step EUV vs. 34-step DUV comparison at 7nm).

[6] Digitimes Asia, "Huawei Ascend 910C Reportedly Hits 40% Yield, Turns Profitable; Aims for 60% Industry Standard" (February 26, 2025); TrendForce, "China Advances in AI Chips: Huawei Reportedly Boosts Yield to 40%, Making Ascend Line Profitable" (February 25, 2025); Reuters, cited in multiple trade sources, estimating 7nm yield below 50% as of mid-2024.

[7] Teddy Bunker and Brent Skorup, "The Lithography Loophole: How China Is Printing Its Way to Chip Self-Sufficiency," *American Enterprise Institute* (April 21, 2026).

[8] Financial Times, cited in Asia Times, "SMIC to Sell Huawei Costly, Inefficient 5nm Chips" (February 14, 2024).

[9] Asia Times (February 14, 2024); American Enterprise Institute (April 21, 2026).

[10] AnySilicon, "Global Foundry Earnings Q3 2025: Growth Diverges as AI, Specialty Nodes and 2nm Ramps Reshape the Market" (2025).

[11] Tom's Hardware, "China to Increase Leading-Edge Chip Output by 5x in Two Years, Report Claims" (February 25, 2026) (citing UBS client note).

[12] SemiAnalysis, "Huawei Ascend Production Ramp: Die Banks, TSMC Continued Production, HBM is The Bottleneck" (February 4, 2026); Tom's Hardware, "China's Chip Champions Ramp Up Production of AI Accelerators at Domestic Fabs, but HBM and Fab Production Capacity Are Towering Bottlenecks" (September 10, 2025).

[13] Tom's Hardware (September 10, 2025); AI Frontiers, "High-Bandwidth Memory: The Critical Gaps in US Export Controls" (February 2, 2026).

[14] Reuters, cited in RCRWireless, "Huawei to Double Output of Ascend AI Chips" (February 25, 2026).

[15] Tom's Hardware, "Chinese Semiconductor Industry Gears Up for Domestic HBM3 Production by the End of 2026" (January 21, 2026); The Substrate, "Where China's AI Chip Supply Chain Stands in 2026" (June 2026).

[16] SemiAnalysis (February 4, 2026); Merics, "Despite Huawei's Progress, Nvidia Continues to Dominate AI Chips Market in China" (March 20, 2025).

[17] Merics (March 20, 2025).

[18] Digitimes Asia (February 26, 2025); ICSmart and Tom's Hardware analyses cited in TrendForce (February 25, 2025).

[19] U.S. Department of Justice, "U.S. Authorities Shut Down Major China-Linked AI Tech Smuggling Network" (December 9, 2025); CNBC, "Nvidia Chips: Plots to Send GPUs to China Expose $160 Million Export-Evasion Web" (December 9, 2025).

[20] Foundation for Defense of Democracies, "Exposure of Major Chinese-Linked Chip Smuggling Operations Shows Limits of Industry Self-Policing" (March 20, 2026); Fortune, "Encrypted Texts Reveal How Nvidia Chips and U.S. Tech Are Being Smuggled to China and Russia" (May 13, 2026).

[21] Foundation for Defense of Democracies (March 20, 2026).

[22] Ray Wang, SemiAnalysis, quoted in CNBC (December 31, 2025).

[23] TrendForce (February 25, 2025); Digitimes Asia (February 26, 2025).

[24] WCCFTech, "China's Premiere Memory Maker Hits HBM3 Roadblock as Domestic DRAM Makers Rush HBM Manufacturing" (April 21, 2026).

[25] The Substrate (June 2026); Tom's Hardware (January 21, 2026).

[26] American Enterprise Institute (April 21, 2026).

[27] TechPowerUp (December 11, 2025) (noting reports of a Chinese domestic EUV prototype undergoing testing as of December 2025).